

Demarcus Williams DSW9740

$$\textcircled{1} \text{ Gain} = \frac{V_o}{V_2 - V_1} = \left(1 + \frac{2R_1}{R_3}\right) \frac{R_3}{R_2}$$

$$\left(1 + \frac{2R_1}{R_3}\right) \frac{R_3}{R_2} = 50$$

$$1 + \frac{2R_1}{R_3} = 50 \Rightarrow \frac{2R_1}{R_3} = 49$$

$$R_1 = 24.5 R_3$$

$$\frac{R_3}{R_2} = 1 \Rightarrow R_2 = R_3$$

$$\therefore \text{If } R_3 = 20 \Omega$$

$$R_1 = 490 \Omega$$

$$R_2 = 20 \Omega = R_3$$

$$b. \left(1 + \frac{2R_1}{R_3} \right) \frac{R_3}{R_2} = A_v$$

$$\frac{R_2}{R_3} A_v = 1 + \frac{2R_1}{R_3}$$

$$\frac{R_2 A_v - R_3}{R_3} = \frac{2R_1}{R_3}$$

$$\frac{R_2 A_v - R_3}{2R_1 R_3} = \frac{1}{R_3}$$

$$\Rightarrow R_3 = \frac{2R_1 R_3}{R_2 A_v - R_3}$$

$$R_3 = \frac{2(20 \times 490)}{20 A_v - 20}$$

$$= 980$$

$$A_v - 1$$

Gain V/V	$R_g (\Omega)$
20	51.6
40	25.1
60	16.6
80	12.4
100	9.89
120	8.23
140	7.05
160	6.16
180	5.47
200	4.92